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Prof. Dr. Detlef Schoder
Cologne Institute for Information Systems (CIIS)
Universität zu Köln
schoder@wim.uni-koeln.de
Ref. Wiss2604-xx

Re: Doctoral Researcher — AI-Driven Protein Design (Ref. Wiss2604-xx)

Dear Prof. Schoder,

I am applying for the doctoral position in AI-driven protein design at CIIS (Ref. Wiss2604-xx). The position's core requirement — integrating physics-informed scoring functions into ML protein design pipelines and building hybrid scoring frameworks that combine AI-derived and physics-based features — is the specific combination I have developed independently across several deployed frameworks. The physics-based scoring is not a planned addition to my toolkit; it is the foundation from which I approach molecular recognition problems.

Physics-informed scoring for protein–protein interactions. The most important result I bring to the project is a zero-parameter method for computing binding affinity from molecular structure. The Crown Prince framework derives dissociation constants as:

$$K_d = \exp(-\Delta M \cdot \ln b)$$

where ΔM is the partition geometry distance between receptor and ligand bounded oscillatory systems, and b is the base of the ternary address encoding. The Hill equation emerges as the exact limiting case of this expression when receptor cooperativity collapses to a single binding mode. Validated against 39 NIST binding compounds at zero free parameters. This is a physics-informed scoring function in the precise sense the position describes: a score derived from molecular geometry rather than from training data, directly integrable into an ML protein design pipeline as a reward signal or as a physics-based feature layer.

For protein–protein interaction prediction specifically, the Syndrome framework applies spectral DFT to three physicochemical channels (hydropathy, volume, charge) to compute a 36-dimensional cosine similarity ρ between any two molecular sequences. Binding is predicted when $\rho \geq \rho^* = 0.72$; the binding ball radius in spectral space is $r^* = \sqrt{2(1-\rho^*)} = 0.748$. Validated: MHC-I/II binding prediction AUC $= 0.81$ vs. NetMHCpan $= 0.68$ on 2,847 IEDB peptides ($r = 0.73$, $p < 10^{-12}$). The spectral complementarity ρ is a physically grounded similarity measure — not a learned distance function — that can be used directly as a physics-based feature in a hybrid scoring framework, alongside GNN-derived or AlphaFold-derived structural features. For PPI lead optimization, the escape predictor tool screens all single amino acid substitutions of a candidate binder for the variant that maximises ρ against the target while minimising off-target complementarity — a systematic computational lead optimization pass in $\mathcal{O}(cL \log L)$.

ML pipeline architecture for protein design. The Borgia framework generates molecular features — electronic transitions, absorption/emission spectra, vibrational modes, and partition coordinates — from molecular structure with zero empirical parameters, via the Triple Equivalence Theorem (oscillatory dynamics = categorical state enumeration = partition operations). These features serve directly as physics-informed input representations to a protein design ML pipeline: instead of learning atomic embedding vectors from a training set, the model receives features derived from molecular physics, which generalise to out-of-distribution protein sequences by construction. The homo-veloce surrogate model system demonstrates the integration pattern in practice: physics-based solvers (Gaussian processes, GNNs, ODE integrators) distilled into lightweight 1D CNNs, LSTMs, and Transformers for real-time inference. This is the same pattern as training an ML protein design model on physics-informed features: the physics ensures structural validity; the ML provides the generative and optimisation capability. The Backward Training Principle formalises the training strategy: a binder model trained at the *pure-intent regime* — the minimum binding energy configuration under maximum on-target specificity, with no solubility or stability constraints — is ϵ-competent at every constrained design sub-task (solubility, stability, off-target avoidance) by feasibility inclusion, not generalisation. A Forward Training Collapse Theorem establishes a positive VC failure gap for the conventional approach of training on filtered feasible designs; a Sample Complexity Theorem gives $(N+1)\times$ efficiency gains for a cascade of N design constraints. This directly addresses the sample efficiency bottleneck in protein design benchmarks.

Structural biology and protein dynamics. The Shakespear platform provides protein conformation, trajectory, catalysis, and dynamics analysis via Kuramoto oscillator synchronization in S-entropy coordinate space. Protein conformational states are represented as bounded oscillatory systems with S-entropy coordinates $(S_k, S_t, S_e) \in [0,1]^3$; trajectory completion reconstructs the full folding pathway from sparse observations using a Banach contraction operator, without forward simulation. For protein design benchmarks, this provides a lightweight computational model for evaluating whether a designed binder adopts the predicted fold — before running full-scale MD or AlphaFold prediction.

Agentic AI and ML pipeline engineering. The position lists willingness to engage with agentic AI approaches. I have built production agentic systems: the Four-Sided-Triangle 8-stage metacognitive RAG pipeline with Rust backend (10–50$\times$ speedup), the Bloodhound distributed categorical VM with Triangle language (navigate/slice/complete), and the Zangalewa Minimum Sufficient Interceptor for $O(\log_k N)$ specialist cascade routing. For protein design, agentic architectures are relevant at the experimental design layer: an agent that iterates between sequence proposal (generative ML), physics scoring (Crown Prince K_d, spectral ρ), structural validation (Shakespear trajectory completion), and experimental feedback is the pipeline the position describes.

Background. I hold an MSc in Computational Biology (Universität Tübingen) and conducted doctoral research in Computational Lipidomics at TUM. I am legally resident in Germany with full work authorisation. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye